

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0509

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting .*

Assessment Tool Weightage: 20%

Scenario-Based Questions

1. Salt-and-Pepper Noise

A scanned document contains salt-and-pepper noise affecting readability.

(a) Explain the scenario and characteristics of this noise.

A scanned document often suffers from Salt-and-Pepper noise, which appears as randomly distributed black (pepper) and white (salt) pixels over the image. This type of noise is also called impulse noise because it affects only a few pixels randomly instead of the entire image. It usually occurs due to faulty camera sensors, dust on the scanner, transmission errors, or hardware malfunctions during image acquisition. As a result, the document becomes difficult to read, and image processing tasks such as OCR (Optical Character Recognition) become less accurate.

Characteristics:

- Appears as random black and white dots.
- Affects only some pixels.
- Caused by sensor or transmission errors.

- Reduces image quality and readability.
- Common in scanned documents and digital images.

(b) Identify key issues impacting image quality.

The main issues caused by Salt-and-Pepper noise are:

- Random black and white pixels reduce image clarity.
- Text becomes difficult to read.
- Fine details and characters may be hidden.
- Image segmentation and edge detection become inaccurate.
- OCR systems may produce incorrect results.
- Overall image quality decreases, affecting further processing.

(c) Apply an appropriate filtering technique.

The most suitable filtering technique is the **Median Filter**.

Input Window (3×3)

10 10 10

10 255 10

10 10 10

↓

Arrange Values

10, 10, 10, 10, 10, 10, 10, 10, 255

↓

Median = 10

Output Window

10 10 10

10 10 10

10 10 10

Working of Median Filter:

1. Select a small neighbourhood (such as a 3×3 window) around each pixel.
2. Arrange all pixel values in ascending order.

3. Find the median (middle) value.
4. Replace the centre pixel with the median value.
5. Repeat this process for every pixel in the image.

This method removes noisy pixels while preserving important edges and details.

(d) Justify why this method is more suitable than alternatives.

The Median Filter is preferred over other filters because:

- It is specifically designed to remove Salt-and-Pepper (impulse) noise.
- It preserves image edges better than Mean or Gaussian filters.
- It removes isolated noisy pixels without causing excessive blurring.
- It improves document readability.
- It provides better results for scanned text images and medical images where edge preservation is important.

Therefore, the Median Filter is considered the most effective solution for this type of noise.

(e) Present your response clearly and logically.

Salt-and-Pepper noise introduces random black and white pixels that reduce the readability and quality of scanned documents. This noise affects text recognition and image analysis. The Median Filter is the most appropriate technique because it effectively removes impulse noise while preserving edges and fine details. After filtering, the image becomes clearer, making it suitable for OCR, segmentation, and further image processing tasks. Hence, the Median Filter is the best choice for improving the quality of images affected by Salt-and-Pepper noise.

2. Object Boundary Detection

In a medical image, boundaries between tissues need to be clearly identified.

(a) Interpret the scenario & explain the importance of boundary detection.

In medical imaging, such as MRI, CT scan, or X-ray, it is important to identify the boundaries between different tissues and organs. Boundary detection helps distinguish normal tissues from abnormal regions such as tumours, cysts, or

lesions. Accurate boundary detection enables doctors to diagnose diseases, measure the size of affected areas, and plan treatments more effectively.

Importance:

- Separates different tissues and organs.
- Detects tumours or abnormal regions.
- Improves medical diagnosis.
- Assists in image segmentation and feature extraction.
- Supports treatment planning and surgical guidance.

(b) Identify key challenges in detecting boundaries.

The major challenges are:

- Low contrast between adjacent tissues.
- Presence of noise in medical images.
- Blurred or weak edges.
- Complex anatomical structures.
- Variation in brightness and intensity.
- Overlapping tissues causing unclear boundaries.

(c) Apply a suitable edge detection method.

The most suitable method is the Canny Edge Detection Algorithm.

Working:

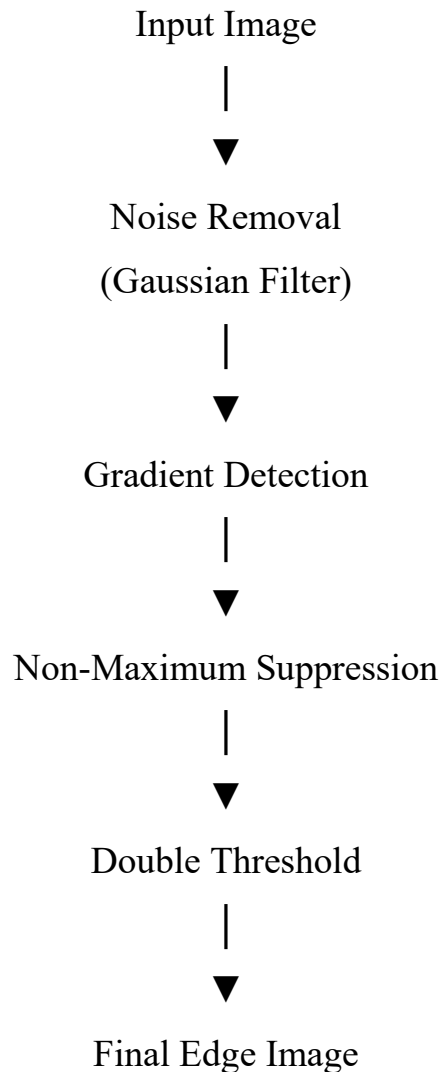
1. Smooth the image using a Gaussian filter.
2. Calculate intensity gradients.
3. Detect edge pixels.
4. Apply non-maximum suppression to remove unwanted edges.
5. Use double threshold and edge tracking to obtain final edges.

(d) Justify your selection with logical reasoning.

The Canny Edge Detection method is preferred because:

- Produces accurate and thin edges.
- Removes noise before edge detection.
- Detects weak and strong edges effectively.

- Reduces false edge detection.
- Provides better performance than Sobel & Prewitt in medical images.



Therefore, Canny Edge Detection is widely used for medical image analysis.

(e) Present your explanation clearly and systematically.

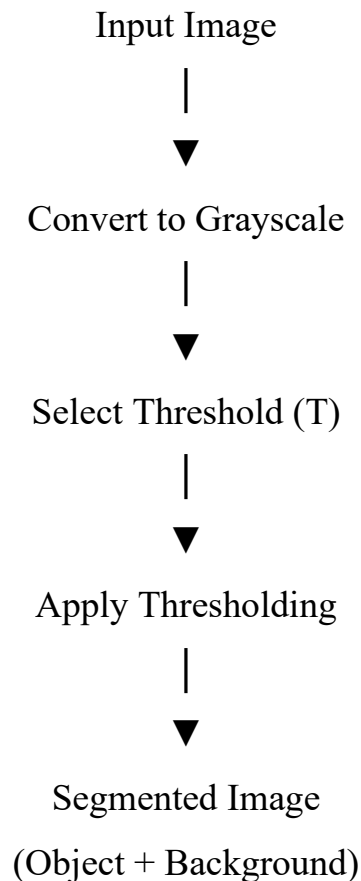
Object boundary detection is essential in medical imaging to identify different tissues and abnormal regions accurately. Among various edge detection techniques, the Canny Edge Detection algorithm provides the best results because it effectively removes noise, detects precise boundaries, and preserves important image details. Hence, it is the most suitable method for detecting tissue boundaries in medical images.

3. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity.

(a) Interpret the scenario and explain segmentation requirements.

Image segmentation is the process of dividing an image into meaningful regions. In this scenario, the objects are clearly distinguishable from the background based on their intensity values.



The objective is to separate the foreground objects from the background so that they can be analysed individually. Segmentation is widely used in medical imaging, industrial inspection, object recognition, and computer vision applications.

Requirements of Segmentation:

- Separate the object from the background.
- Group similar pixels together.
- Produce accurate object boundaries.
- Simplify image analysis.

- Improve feature extraction and object recognition.

(b) Identify key features enabling separation.

The important features that help separate the object from the background are:

- Difference in pixel intensity.
- High contrast between object and background.
- Uniform object region.
- Distinct edges.
- Similar grayscale values within the object.

These characteristics make segmentation easier and more accurate.

(c) Apply a suitable segmentation technique.

The most suitable technique is Threshold Segmentation (Thresholding).

Working:

1. Convert the image to grayscale (if required).
2. Select a threshold value (T).
3. Compare each pixel with the threshold.
4. If the pixel intensity is greater than T , classify it as an object.
5. Otherwise, classify it as the background.

This method separates the image into two regions: foreground and background.

(d) Justify your decision logically.

Threshold Segmentation is the best choice because:

- It is simple and easy to implement.
- Works well when the object and background have different intensity values.
- Requires less computation time.
- Produces fast and accurate segmentation.
- Suitable for images with clear contrast.

Hence, Threshold Segmentation is widely used for simple image segmentation tasks.

(e) Present your response clearly.

Simple segmentation divides an image into meaningful regions by separating the object from the background. Since the objects have different intensity values from the background, Threshold Segmentation is the most suitable method. It is simple, fast, and provides accurate results when there is a clear intensity difference. This technique improves object detection and simplifies further image analysis.

4. Low Brightness Image

An image captured in low light appears very dark with poor visibility.

(a) Interpret the scenario and explain the cause of low brightness.

A low-brightness image is captured under poor lighting conditions, making it appear dark and unclear. This can occur due to insufficient light, low camera exposure, improper camera settings, or shadows. As a result, important objects and details become difficult to identify, reducing the quality of the image.

Causes of Low Brightness:

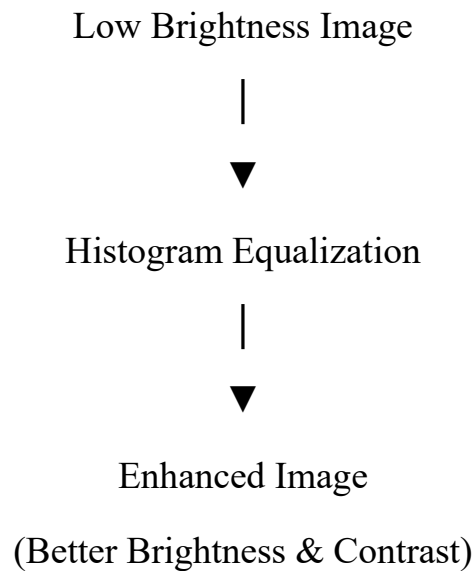
- Poor lighting conditions.
- Low camera exposure.
- Incorrect camera settings.
- Shadows or uneven illumination.
- Low sensor sensitivity.

(b) Identify key issues affecting image visibility.

The major issues are:

- Poor visibility of objects.
- Loss of important details.
- Low contrast between object and background.

- Difficulty in object detection.
- Reduced accuracy in image analysis.
- Eye strain while viewing the image.



(c) Apply an appropriate enhancement technique.

The most suitable technique is Histogram Equalization.

Working:

1. Read the input image.
2. Convert it to grayscale (if required).
3. Calculate the histogram of pixel intensities.
4. Redistribute the intensity values evenly.
5. Generate an enhanced image with improved brightness and contrast.

(d) Justify your choice with reasoning.

Histogram Equalization is preferred because:

- It improves image brightness and contrast.

- Enhances hidden details in dark regions.
- Spreads pixel intensity values uniformly.
- Simple and computationally efficient.
- Better than simple brightness adjustment, as it enhances the overall image quality instead of only increasing brightness.

Therefore, Histogram Equalization is widely used for enhancing low-light images.

(e) Present your explanation clearly.

Low-brightness images reduce visibility and make object identification difficult. **Histogram Equalization** enhances the image by improving brightness and contrast, making hidden details more visible. It is an effective and widely used image enhancement technique for low-light conditions, improving both image quality and analysis accuracy.

5. Gaussian Noise

An image contains Gaussian noise affecting smooth regions.

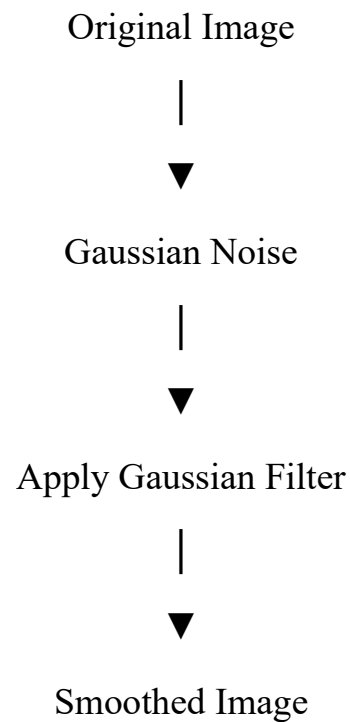
(a) Interpret the scenario and describe Gaussian noise.

Gaussian noise is a type of random noise that affects the entire image. It is caused by camera sensors, electronic circuits, or poor lighting conditions during image acquisition. This noise follows a Gaussian (Normal) distribution, making the image appear grainy and reducing its overall quality. It mainly affects smooth regions, making image analysis difficult.

Characteristics:

- Randomly distributed over the image.
- Follows a Gaussian (Normal) distribution.

- Makes the image appear grainy.



(b) Identify key issues affecting quality.

The major issues caused by Gaussian noise are:

- Image becomes grainy and unclear.
- Fine details are hidden.
- Reduced contrast and sharpness.
- Edge detection becomes difficult.
- Decreased accuracy in image processing.
- Lower visual quality.

(c) Apply an appropriate smoothing filter.

The most suitable smoothing filter is the Gaussian Filter.

Working:

1. Read the noisy image.
2. Create a Gaussian kernel (e.g., 3×3 or 5×5).
3. Convolve the kernel with the image.
4. Smooth the noisy pixels.
5. Obtain a cleaner and clearer image.

(d) Justify your method over alternatives.

The Gaussian Filter is preferred because:

- It is specially designed to reduce Gaussian noise.
- Smooths the image while preserving important structures.
- Produces natural-looking results.
- More effective than the Mean Filter for Gaussian noise.
- Improves image quality for further processing.

Hence, the Gaussian Filter is the best choice for images affected by Gaussian noise.

(e) Present your explanation clearly.

Gaussian noise is a common type of noise that degrades image quality by introducing random intensity variations. Applying a **Gaussian Filter** effectively smooths the image, reduces noise, and improves clarity while preserving important image information. Therefore, it is the most suitable enhancement technique for Gaussian noise.

6. Loss of Fine Details

An image loses fine details after processing.

(a) Interpret the scenario and explain detail loss.

Fine details in an image may be lost after image processing due to excessive smoothing or filtering. This causes small textures, edges, and important features to disappear, making the image look blurred. Detail loss reduces image quality and affects tasks such as object recognition and medical diagnosis.

Causes of Detail Loss:

- Excessive smoothing.
- Improper filtering.
- Low-resolution images.
- Compression effects.
- Noise removal using large filters.

(b) Identify key issues related to filtering.

The major issues are:

- Loss of edges and fine details.
- Blurred image appearance.
- Reduced sharpness.
- Difficulty in feature extraction.
- Lower accuracy in object detection.
- Poor visual quality.

(c) Apply a suitable enhancement technique.

The most suitable technique is Image Sharpening using the Laplacian Filter.

Working:

1. Read the input image.

2. Apply the Laplacian operator.
3. Detect edges and fine details.
4. Add the sharpened result to the original image.
5. Obtain a sharper image with enhanced details.

(d) Justify your method with reasoning.

The Laplacian Filter is preferred because:

- Enhances edges and fine details.
- Improves image sharpness.
- Restores details lost during smoothing.
- Simple and computationally efficient.
- Suitable for medical, satellite, and industrial images.

(e) Present your explanation clearly.

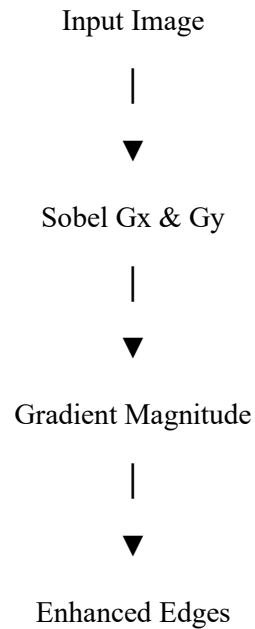
Loss of fine details reduces the quality of an image and makes object recognition difficult. Applying the **Laplacian Filter** enhances edges and restores important details, producing a sharper and clearer image. Therefore, it is an effective enhancement technique for improving images that have lost fine details.

7. Edge Enhancement Requirement

An application requires highlighting edges for feature extraction.

(a) Interpret the scenario and explain the need for edge enhancement.

Edge enhancement is used to make object boundaries more visible. It improves the visibility of edges so that important features can be extracted accurately. This is useful in medical imaging, face recognition, fingerprint analysis, and object detection.



Need for Edge Enhancement:

- Highlights object boundaries.
- Improves feature extraction.
- Helps in object recognition.
- Increases image clarity.
- Supports segmentation and analysis.

(b) Identify key issues in detecting edges.

The main challenges are:

- Noise creates false edges.
- Weak or blurred edges.
- Low image contrast.
- Uneven illumination.
- Complex object shapes.

(c) Apply an appropriate method.

The most suitable method is the Sobel Edge Detection Method.

Working:

1. Read the input image.
2. Convert it to grayscale.
3. Apply Sobel horizontal (Gx) and vertical (Gy) filters.
4. Calculate the edge magnitude.
5. Display the enhanced edge image.

(d) Justify your choice logically.

The Sobel Operator is preferred because:

- Simple and easy to implement.
- Detects both horizontal and vertical edges.
- Enhances important object boundaries.
- Fast and computationally efficient.
- Suitable for feature extraction in computer vision applications.

(e) Present your explanation clearly.

Edge enhancement improves the visibility of object boundaries, making feature extraction more accurate. The **Sobel Operator** detects horizontal and vertical edges efficiently while maintaining low computational complexity. Hence, it is a suitable technique for highlighting edges in computer vision applications.

Code of Conduct:

I B. Siva Parvathi (192525194) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B. Siva Parvathi

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand